Project Proposal (Proposed Solution) Report

Date	30 September 2026
Team ID	SWTID-2026-3818
Project Name	Auto_Ticket_Classification_Using_Flow_Diagram
Maximum Marks	5 Marks

The proposal report aims to transform ticket handling using machine learning, improving speed and accuracy. Key features are an ML-based ticket classifier, priority prediction, automatic routing and a flow diagram for every ticket.

Project Overview

Objective	Build a web application that automatically classifies IT support tickets into categories, predicts priority, routes them to the right team and visualises the process with a flow diagram.
Scope	Text-based support tickets in English. The system covers ticket intake, preprocessing, classification, routing, flow diagram generation, agent feedback and an admin dashboard.

Problem Statement

Description	Manual sorting of support tickets is slow, inconsistent and hard to track, which delays resolution and lowers user satisfaction.
Impact	Solving this will reduce response time, lower the workload of agents, improve routing accuracy and give managers a clear view of how tickets flow.

Proposed Solution

Approach	Use Natural Language Processing (TF-IDF) with a supervised machine learning classifier trained on labelled tickets. A confidence threshold decides whether a ticket is routed automatically or sent for manual review. A Graphviz flow diagram is generated for every ticket.
Key Features	• ML-based category prediction with confidence score • Priority prediction (High / Medium / Low) • Automatic routing to the responsible team • Per-ticket flow diagram • Agent feedback and retraining • Admin dashboard and reports

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop CPU (Intel i3/i5 or above, 4 cores); GPU not required
Memory	RAM specifications	8 GB
Storage	Disk space for data, models and logs	256 GB SSD (project needs < 1 GB)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, nltk, joblib, graphviz, matplotlib, seaborn

Resource Type	Description	Specification / Allocation
Development Environment	IDE	VS Code, Jupyter Notebook / Google Colab
Data		
Data	Source, size, format	[Dataset name and source, e.g. Kaggle IT support tickets], [number of rows], CSV

Proposed Workflow (Flow Diagram)

